ChE 2O04 Tutorial 2 Solutions

Tutorial Problem #1: In the table below are shear stress and shear rate data for three bottles of syrup (simply identified as A, B, & C). One bottle is supposed to contain pure grade 'Canada #1 very clear' maple syrup (which as we discussed in lecture is a Newtonian fluid) while the other two are thought to be cheaper grade 'pancake syrups' (i.e., corn syrup plus colouring and other additives) that have some non-Newtonian rheological behaviour.

a. Analyze the data using the linear regression function in Excel to determine which bottle (if any) contains the 'pure' maple syrup; be sure to give a clear description of the exact reasons how you conclude that the syrup is a Newtonian or non-Newtonian fluid.

b. For the pure maple syrup, calculate the viscosity (in cP units).

Bottle A		Bottle B		Bottle C	
Shear Rate (1/s)	Shear Stress (Pa)	Shear Rate (1/s)	Shear Stress (Pa)	Shear Rate (1/s)	Shear Stress (Pa)
80	45	80	56	80	51
120	71	120	77	120	72
160	98	160	101	160	99
200	124	200	126	200	121
240	148	240	152	240	148
280	173	280	175	280	171
320	195	320	200	320	197
360	216	360	226	360	218
400	238	400	250	400	246

Solution to Tutorial Problem #1:

Bottle A SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.999054849
R Square	0.998110592
Adjusted R Square	0.997840677
Standard Error	3.078342164
Observations	9

Interpretation: The residual plot shows a clear parabolic trend. Thus, a linear model here is not appropriate thus indicating a Non-Newtonian fluid. Because there is a clear pattern in the residuals, we **DO NOT** need to look at the 95% CI values (highlighted in yellow)

ANOVA

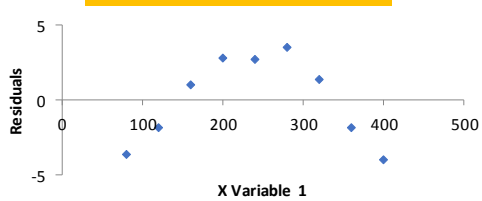
	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	35041.66667	35041.67	3697.864	8.53869E-11
Residual	7	66.33333333	9.47619		
Total	8	35108			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	0.333333333	2.595886041	0.128408	0.901437	-5.804961754	6.471628421	-5.80496175	6.471628421
X Variable 1	0.604166667	0.009935307	60.81007	8.54E-11	0.5806734	0.627659934	0.5806734	0.627659934

RESIDUAL OUTPUT

Observation	*Predicted Y*	*Residuals*
1	48.66666667	-3.666666667
2	72.83333333	-1.833333333
3	97	1
4	121.1666667	2.833333333
5	145.3333333	2.666666667
6	169.5	3.5
7	193.6666667	1.333333333
8	217.8333333	-1.833333333
9	242	-4

Bottle A Residual Plot



Bottle B SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.999816485
R Square	0.999633004
Adjusted R Square	0.999580576
Standard Error	1.374368542
Observations	9

Interpretation: Although a linear model is appropriate here (random scatter in residual plot), this model also has a non-zero intercept and is therefore non-Newtonian. This fluid most likely behaves similarly to Bingham plastic. (Non-zero intercept + positive slope).

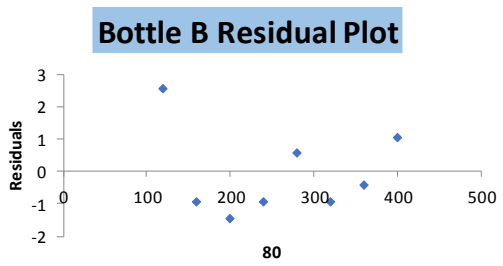
ANOVA

	df	SS	MS	F	Significance F
Regression	1	36015	36015	19066.76471	2.75619E-13
Residual	7	13.22222222	1.888888889		
Total	8	36028.22222			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	4.444444444	1.158969316	3.834824944	0.006417482	1.703917492	7.184971397	1.703917492	7.184971397
X Variable 1	0.6125	0.004435755	138.0824562	2.75619E-13	0.602011105	0.622988895	0.602011105	0.622988895

RESIDUAL OUTPUT

Observation	Predicted Y	Residuals
1	53.44444444	2.555555556
2	77.94444444	-0.944444444
3	102.4444444	-1.444444444
4	126.9444444	-0.944444444
5	151.4444444	0.555555556
6	175.9444444	-0.944444444
7	200.4444444	-0.444444444
8	224.9444444	1.055555556
9	249.4444444	0.555555556



Bottle C SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.999742552
R Square	0.999485171
Adjusted R Square	0.999411624
Standard Error	1.621286967
Observations	9

Interpretation: Here, the confidence interval for the intercept term contains 0 and thus a line with a slope of 0.61 stems from the origin as the linear model of the data. The residual plot shows a random scatter and thus a linear model is appropriate. This fluid is Newtonian and most likely the pure grade Maple syrup. The viscosity is the slope of the model, 0.61 Pa*s or 610cP.

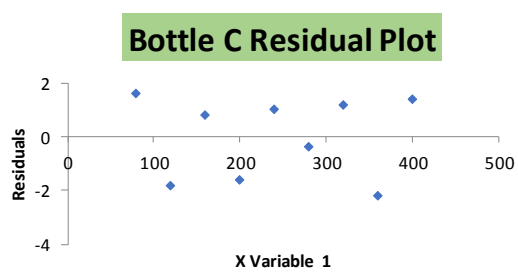
ANOVA

	df	SS	MS	F	Significance F
Regression	1	35721.6	35721.6	13589.73913	9.0124E-13
Residual	7	18.4	2.628571429		
Total	8	35740			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.6	1.367189215	0.438856592	0.673991943	-2.632888774	3.832888774	-2.632888774	3.832888774
X Variable 1	0.61	0.005232681	116.5750365	9.0124E-13	0.597626675	0.622373325	0.597626675	0.622373325

RESIDUAL OUTPUT

Observation	Predicted Y	Residuals
1	49.4	1.6
2	73.8	-1.8
3	98.2	0.8
4	122.6	-1.6
5	147	1
6	171.4	-0.4
7	195.8	1.2
8	220.2	-2.2
9	244.6	1.4

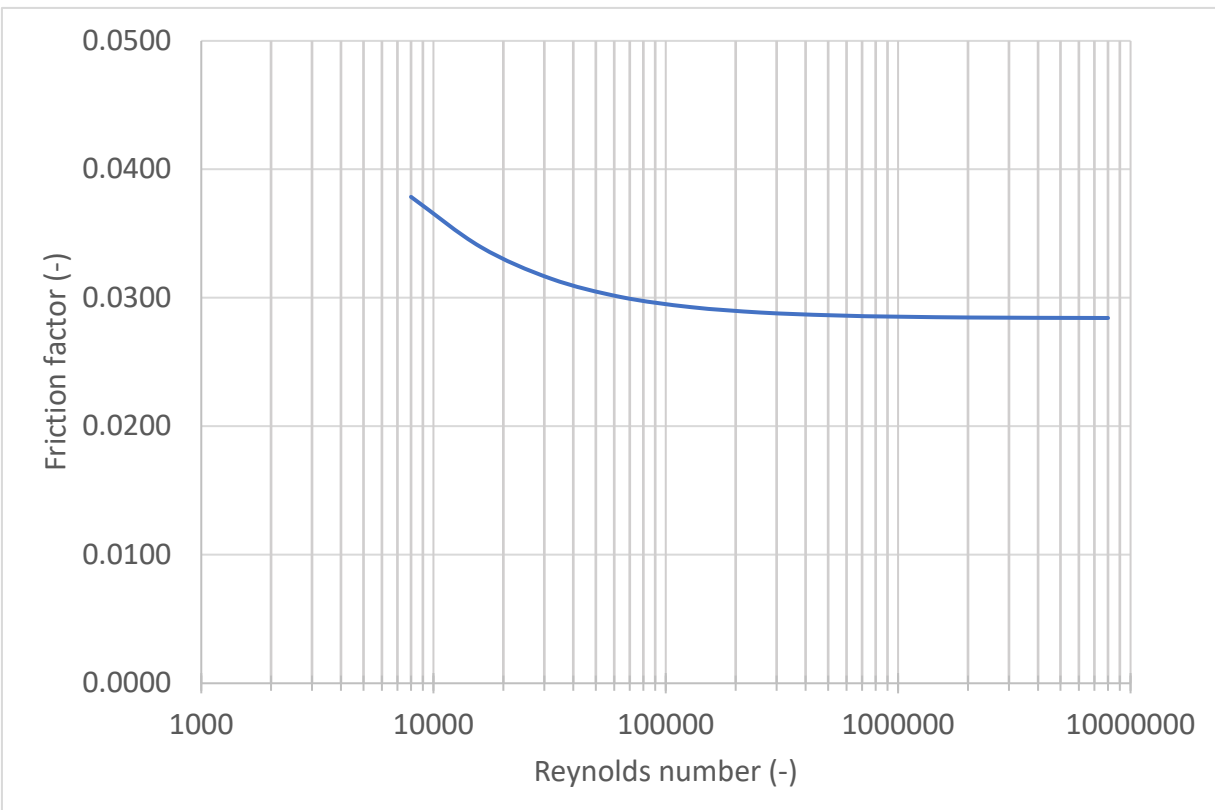


Tutorial Problem 2

Relative roughness 0.004

Reynolds number	LHS of Colebrook Equation	Friction Factor
8000	2.87394E-07	0.0379
16000	2.44959E-07	0.0339
32000	2.17223E-07	0.0315
64000	-2.1752E-07	0.0301
128000	3.27692E-08	0.0293
256000	-1.53797E-08	0.0289
512000	5.31479E-08	0.0286
1024000	5.02499E-09	0.0285
2048000	3.78247E-10	0.0285
4096000	-5.87627E-09	0.0284
8000000	-8.20936E-10	0.0284

Notice how the friction factor barely changes once the Reynolds number gets up above 250,000



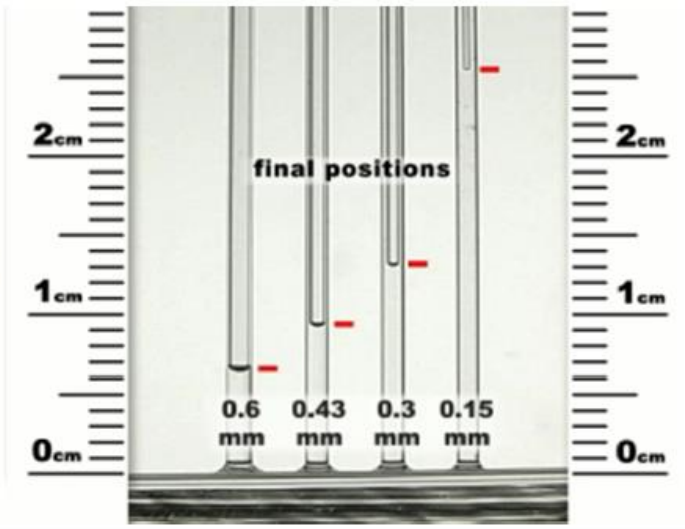
Tutorial Problem 3

a. Create a nice figure in Excel that shows the profile for height of water inside a capillary tube varies with the radius of the capillary tube (for range 0.1 to 5 mm) as determined by:

$$\Delta h = \frac{2\sigma \cos\theta}{\rho g r_{capillary}}$$

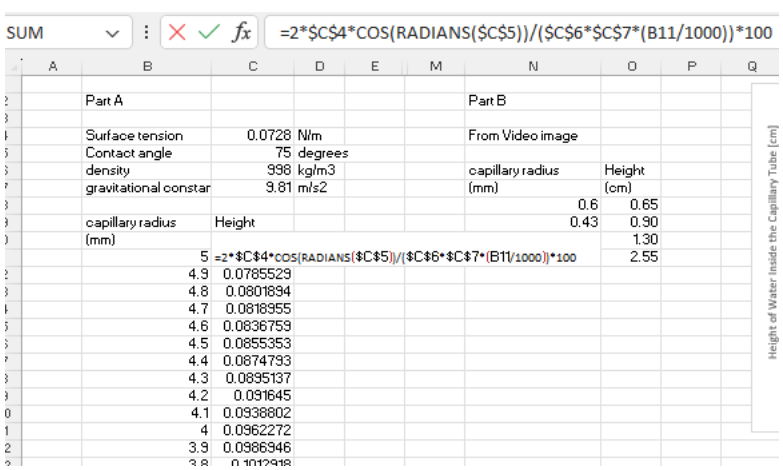
when $\sigma = 0.0728$ N/m
$\theta = 75°$
$\rho = 998$ kg/m³

b. Make a new figure that has the profile from part a) along with the four experimental values shown in the image to the right. Then provide a short commentary on the overall agreement between the predicted values (from equation in part a) and the experimental values.



Solution Process:

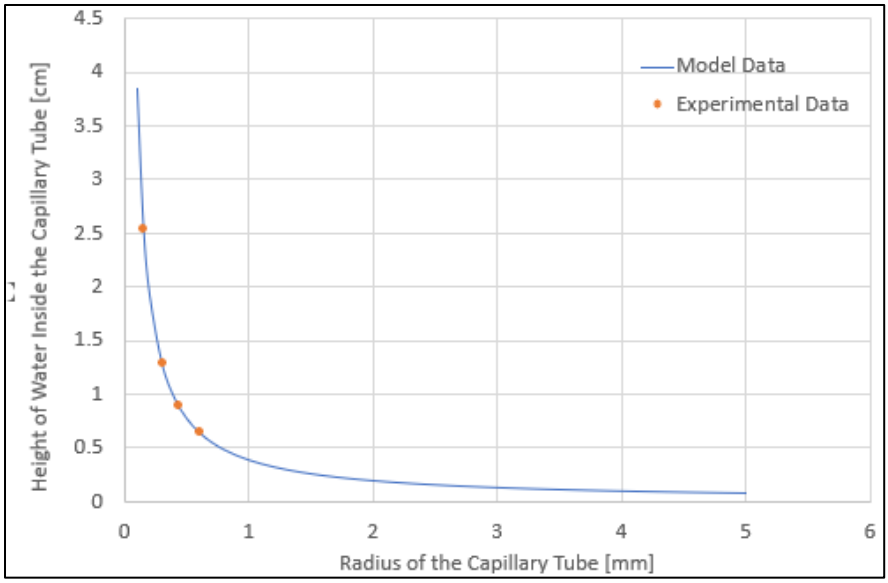
- In this question, you are plotting experimental data against model data that will be created in Excel based on the equation given above. Make sure to include your constants in the Excel, and reference them for easier modification. The model data can be created, as shown below:



SUM | =2*C4*COS(RADIANS(C5))/(C6*C7*(B11/1000))*100

	A	B	C	D	E	M	N	O	P	Q
2		Part A					Part B			
3										
4		Surface tension	0.0728	N/m			From Video image			
5		Contact angle	75	degrees						
6		density	998	kg/m3			capillary radius	Height		
7		gravitational constar	9.81	m/s2			(mm)	(cm)		
8							0.6	0.65		
9		capillary radius	Height				0.43	0.90		
10		(mm)						1.30		
		5	=2*C4*COS(RADIANS(C5))/(C6*C7*(B11/1000))*100					2.55		
12		4.9	0.0785529							
13		4.8	0.0801894							
14		4.7	0.0818955							
15		4.6	0.0836759							
16		4.5	0.0855353							
17		4.4	0.0874793							
18		4.3	0.0895137							
19		4.2	0.091645							
20		4.1	0.0938802							
21		4	0.0962272							
22		3.9	0.0986946							
23		3.8	0.1012918							

- Once the model data has been created, the model data can be plotted against the experimental data given the image in the solution prompt. So general points for the plot are:
  - Include relevant and informative axis titles including the units, and informative legends
  - Model data can be included as a line plot, experimental data should be included as a scatter plot.



Visually, we can see good agreement between the model and experimental data in this question. If we looked at the difference between the experimental and model values at each of the datapoints, we can see the model error does not significantly increase at smaller radii.

From Video image		
capillary radius (mm)	Height (cm)	Diff between experimental and model values (cm) (cm)
0.6	0.65	0.008
0.43	0.90	0.005
0.3	1.30	0.017
0.15	2.55	-0.016